

20261007 Distribution of NSF BPT Distances Across Infall Stages

Let $x = \log_{10}([\text{NII}]/\text{H}\alpha)$ or $\log_{10}([\text{SII}]6716 + [\text{SII}]6730)/\text{H}\alpha$, and $y = \log_{10}([\text{OIII}]/\text{H}\beta)$. Here i use the KE01 demarcation curve:

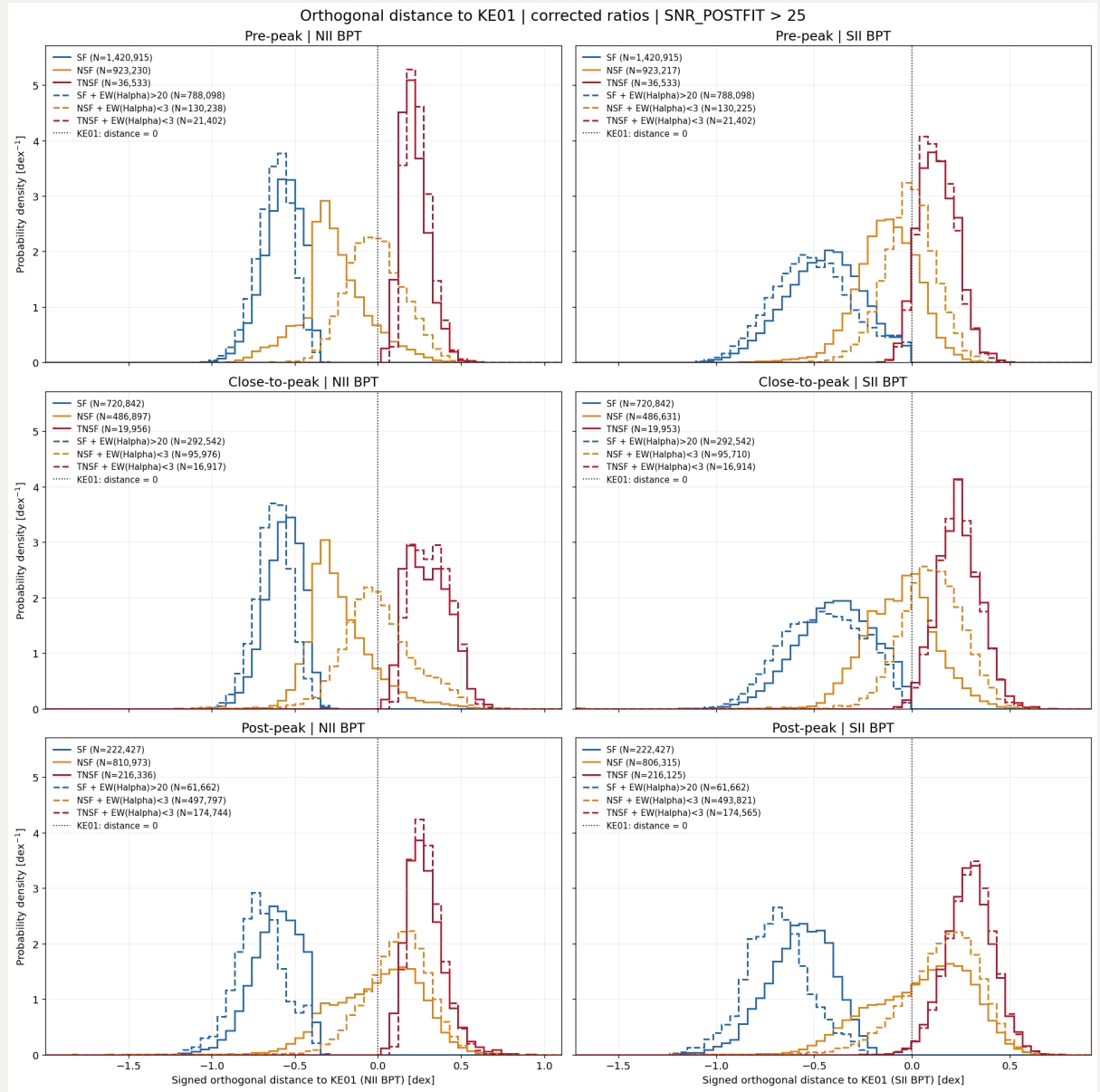
$$\begin{aligned} f_{\text{NII}}(u) &= \frac{0.61}{u - 0.47} + 1.19, & u < 0.47, \\ f_{\text{SII}}(u) &= \frac{0.72}{u - 0.32} + 1.30, & u < 0.32. \end{aligned} \tag{1}$$

The orthogonal magnitude is the shortest Euclidean distance to the curve in log-ratio space (with $b = 0.47$ for NII, $b = 0.32$ for SII):

$$|d_{\perp}| = \min_{u < b} \sqrt{(x - u)^2 + [y - f(u)]^2}. \tag{2}$$

Here we still have 6 different classifications, but for the main purpose, we only look at the SF (solid blue) and NSF (solid organge). Perhaps also the HOLMES components (red).

So a bit surprised to see that close-to-peak distributions are almost identical to pre-peak; while in post-peak, the NSF clearly move to positive side.



And we can also checkout the NSF BPT directly:

